

Final Project Proposal
Dr. Li
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Part 1: Team Information

Team Members:

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Team Responsibilities:

- Jasmina: Abstract, Introduction, Data Exploration/Preprocessing, Model Implementation
- Gianella: Background, Related Work, Conclusion, Results Analysis
- Shared: Experiments, Visualizations, Presentation Preparation

Part 2: Project Description

Project Topic: Credit Card Fraud Detection Using Cost-Sensitive Machine Learning

1. Project Statement:

Credit card fraud poses a significant financial threat, with global losses exceeding \$33 billion annually (The Nilson Report). Our project aims to reduce major financial losses while avoiding unnecessary blocks and false alarm rates on legitimate customers transactions.

Source: <https://finance.yahoo.com/news/global-card-fraud-losses-33-170800960.html>

Objectives:

- Achieve a 60 percent reduction in losses through better detection of fraud.
- The false positive rate should be kept low, below 1 percent, to reduce friction for customers.
- It should create a cost-sensitive model that accounts for the asymmetric costs of error types (false positives and false negatives).
- Determine the best decision levels based on business costs rather than accuracy alone.

2. Datasets:

Primary Dataset: Credit Card Fraud Detection Dataset (Kaggle)

- Source: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- Size: 284,807 transactions collected over 2 days (September 2013)
- Features: 31 attributes
 - V1-V28: PCA-transformed anonymized features (for privacy protection)
 - Time: Seconds elapsed since first transaction
 - Amount: Transaction amount in Euros
 - Class: Binary target variable (0 = legitimate, 1 = fraud)
- Class Distribution: Highly imbalanced - 492 frauds (0.172%) vs. 284,315 legitimate (99.828%)
- Quality: No missing values, pre-normalized PCA features

Alternative Datasets:

- <https://www.kaggle.com/datasets/whenamancodes/fraud-detection>
- <https://www.kaggle.com/datasets/kartik2112/fraud-detection>

3. Data Mining Techniques (Brief Description)

In our project, we will focus on supervised binary classification, with particular attention to handling extreme class imbalance. Since only 0.172 percent of transactions are fraudulent, we will consider other resampling methods, such as SMOTE (Synthetic Minority Over-sampling Technique), undersampling, or oversampling to balance the dataset and improve model performance. We will use a few different classification algorithms, such as Logistic Regression as our baseline, Random Forests, and XGBoost (or other gradient boosting algorithms), that may also combine multiple models to improve performance.

One of the core innovations in our scheme will be cost-sensitive learning, in which we will judge the model based on real business impact, not just accuracy. Missing a fraud transaction costs more money than incorrectly blocking a real transaction, so the model must account for this difference.

To assess it, we will consider metrics suitable for imbalanced data, including Precision, Recall, F1-Score, and Precision-Recall AUC, instead of straight accuracy. Additionally, we will examine the confusion matrix to understand the trade-offs between various types of errors. Through our discoveries, we will conclude new findings and determine which features could lead to fraud being detected such as credit scores, amount limit, and location. We can also consider using anomaly detection or other approaches to improve performance through further research, based on our preliminary findings and ongoing studies.

4. Planned Approaches Summary:

Handling Imbalanced Data: - Resampling techniques (SMOTE, undersampling, or oversampling) - Class weighting adjustments - Threshold optimization for classification	Classification Algorithms: - Logistic Regression (baseline) - Random Forest - XGBoost/Gradient Boosting - Possibly ensemble methods
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